

AIRSTREET · REQUIREMENTS

AirStreet requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

AirStreet requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish and the first co-design partner, and will be checked by calculation at TRL 3. Status reflects the TRL 2 estimates in [02-concept.md](#). Two requirements are **not met** (R3 and R15) and four are at risk.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measure PM _{2.5} at street level	0 to 500 µg/m ³ ; hourly mean after humidity correction within ±5 µg/m ³ or ±30 % (whichever is larger) of a collocated reference	CalRig chamber check, then field collocation	Plausible; field result unverified
R2	Measure NO ₂ at street level	0 to 200 ppb (about 0 to 380 µg/m ³); hourly mean absolute error 5 ppb or less against a reference station after field calibration	Field collocation with a chemiluminescence reference station	At risk; literature suggests about 3.5 ppb is reachable (Zimmerman et al., 2018)
R3	Resolve NO ₂ near the WHO annual guideline	Annual mean uncertainty 2 µg/m ³ or less at 10 µg/m ³	Collocation statistics	Not met. 5 ppb (about 9.4 µg/m ³) hourly error is similar to the guideline itself; long averages help, but drift is unverified
R4	Distinguish streets that exceed the EU NO ₂ limit value	Classify annual means above or below 40 µg/m ³ with 90 % confidence	Collocation statistics	Plausible
R5	Measure temperature and humidity for corrections	±0.5 °C and ±3 % RH inside the shield; radiation error 1 °C or less in full sun at 1 m/s wind	CalRig chamber check; side-by-side field test against an aspirated reference	At risk; shield performance unverified
R6	Calibrate with a stated method	T, RH and PM checked on CalRig before deployment; NO ₂ collocated with a reference station for at least 14 days before deployment and at least every 6 months	Calibration record per node	Met by design
R7	Keep raw data for recalibration	Send raw NO ₂ working and auxiliary electrode signals, PM, T and RH with a calibration version tag	Firmware and data schema review	Met by design
R8	Report often enough for street maps	One record every 5 min; hourly means published within 15 min; 7 days of store and forward	Timing and storage calculation	Met by design (estimate)
R9	Fit the FieldNode power budget	Sensor average 115 mW or less; energy neutral at the FieldNode design sun hours	Power budget calculation	Met, about 45 mW (estimate)
R10	Stay within radio duty-cycle rules	LoRaWAN airtime below 1 % in EU868 sub-bands	Airtime calculation	Met, about 0.06 % at SF9 and 0.5 % at SF12 (estimate)
R11	Mount on street poles without drilling	Poles 80 to 200 mm diameter; band clamps only; inlets between 1.5 and 4 m above ground	Design review	Met by design
R12	Survive outdoors	Pod inlets face down behind insect mesh with drip lid; electronics in	Design review, later field trial	At risk; condensation and insect ingress unverified

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
		the FieldNode IP65 enclosure; -10 to 50 °C		
R13	Light and compact on the pole	Mass 3.5 kg or less; frontal area 0.12 m ² or less	Massing model	Met, about 3.0 kg and 0.09 m ² (estimate)
R14	Collect levels only	No images, audio or personal identifiers collected or sent	Design review	Met by design
R15	Low cost	Parts \$180 or less per node	Priced BOM	Not met. About \$391 with the FieldNode core; about \$265 for the sensor head alone
R16	Serviceable in the field	NO2 sensor and PM sensor replaceable in 15 min at the pole with hand tools	Design review	At risk; access from a ladder unverified

■ Assumptions

- NO2 conversions use 1 ppb = 1.88 µg/m³ at 25 °C and 101.325 kPa.
- The WHO guideline level for annual NO2 is 10 µg/m³ and for PM2.5 is 5 µg/m³, as cited by the European Environment Agency ([EEA, 2024](#)). The EU annual limit value for NO2 is 40 µg/m³ under Directive 2008/50/EC, which also sets sampling inlets between 1.5 m and 4 m above ground ([EUR-Lex](#)).
- FieldNode allows about 115 mW average for sensors (FieldNode README, lab shared components set).
- R1 and R2 targets are proposals for review, set with reference to published field results; they are not taken from a standard.